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Submitter Information

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General Comment

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March 15, 2025

National Science Foundation
2415 Eisenhower Avenue

Subject: Submission in Response to the Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Dear National Science Foundation Representative,

I am pleased to submit my comments in response to the National Science Foundation's Request for Information (RFI) on the Development of an Artificial Intelligence (AI) Action Plan, as published in the Federal Register. As an expert in artificial intelligence, machine learning, clinical decision-making, and data privacy, I have carefully considered the challenges and opportunities presented by the AI revolution. My enclosed document, On the Elements of the AI Action Plan, provides specific recommendations to ensure the responsible development, deployment, and governance of AI technologies for the benefit of society.

In my response, I address key issues such as training data transparency, intellectual property rights, equitable economic models, ecological responsibility, and the necessity for national and international regulatory frameworks. As AI-driven automation increasingly shapes industries and economic structures, it is imperative that policies proactively mitigate risks associated with workforce displacement, wealth concentration, and regulatory gaps.

The recommendations outlined in my submission highlight the urgency of developing enforceable standards for AI accountability, fair compensation for data contributors, and sustainable AI accessibility models. Additionally, I emphasize the necessity of federal legislation to ensure the United States maintains both global leadership and societal resilience in the AI era.

I appreciate the NSF's efforts in gathering expert insights on AI policy and strategy. I hope that my submission contributes meaningfully to the ongoing discourse and informs policy development that fosters an equitable and sustainable AI-driven future.

Please feel free to contact me should you require any further information or clarification regarding my submission.

Sincerely,

Attachments

On the Elements of AI Action Plan - by Mehmet Kayaalp MD PhD - 2025-03-15 - Maryland

On the Elements of the AI Action Plan

by
Mehmet Kayaalp, M.D., Ph.D.

Executive Summary

We are at the dawn of the AI revolution. While it's easy to focus solely on technological progress, overlooking its social implications could create severe long-term consequences. Many countries are closely matched in AI technology, but those that will benefit the most are those that will proactively manage social impacts before they become overwhelming. Current AI policies, including the EU AI Act, are often too vague or unenforceable. Here, I offer specific, actionable recommendations based on my expertise in AI, clinical decision-making, and privacy.

My key proposals include:

- Transparent and auditable AI training processes, especially for high-risk applications.
- Clear documentation of AI systems, ensuring accountability and intellectual property rights protection.
- Equitable compensation for individuals and entities whose data supports AI system development.
- Ecological accountability from AI companies, ensuring they minimize their carbon footprint.
- Strong human-in-the-loop regulations, especially in critical sectors like medicine and security.
- Translating global cultural heritage into accessible datasets to enhance AI models.

To sustain a healthy society and economy, we must address these aspects urgently rather than reactively. The US currently lacks comprehensive federal AI regulations, and it's critical to establish them now for both global leadership and long-term social stability.

Introduction

We stand at the beginning of the AI revolution. Focusing exclusively on technological advancement without adequately addressing the social ramifications could be profoundly detrimental. As we have seen in recent months, other nations are not far behind in the

technological aspects of this revolution. The emergence of models like Deepseek R1 has demonstrated China's prowess in this field, evoking a sentiment akin to the Sputnik moment. However, technological superiority alone will be meaningless if societies fail to proactively adapt to the new era. Nations that will be able to reap the gains from this revolution will be those that proactively adjust their social systems before the adverse effects arrive in tsunamis—at which point, responding may no longer be feasible.

In reviewing international AI policies—including those of the EU and various U.S. states—using AI-assisted methods, I have identified that most existing regulations and frameworks are abstract, high-level, and challenging to enforce practically. Here, I provide concrete, actionable recommendations. However, please interpret numeric examples (e.g., revenue-sharing percentages) as illustrative rather than prescriptive.

My recommendations draw specifically from my professional expertise in AI, machine learning, clinical decision-making, and data privacy.

Training Data and Process Transparency

AI systems' performance heavily depends on training data, which encompasses foundational training, safety tuning, reinforcement learning from human feedback, fine-tuning, and retrieval-augmented generation. Negligent or unethical practices in curating training data can lead to harmful outcomes in critical sectors.

AI developers must adhere to rigorously documented protocols. Deviations from these protocols must be transparently communicated to end-users and regulators. Consequences for undisclosed deviations should align proportionally with domain risk:

- **High-risk** (e.g., homeland security, nuclear controls): Severe penalties.
- **Moderate-risk** (e.g., clinical decision-making, robotic assistance to elderly): Significant penalties.
- **Low-risk** (e.g., word processing, video gaming): Lesser penalties.

For example, in intelligent clinical decision-making, failure to disclose omitted validation steps could jeopardize patient safety and thus warrant significant regulatory action.

Several governments have introduced varying transparency mandates. Colorado[1] and California[2] have high-level requirements for generative AI systems, while the EU[3] mandates comprehensive disclosures, including data provenance, cleaning methods, bias mitigation practices, and validation details.

I recommend combining both approaches. Specifically, for high-risk applications, developers must provide comprehensive documentation—including random seed values—to enable independent replication, especially critical when claims are legally contested.

Intellectual Property (IP) Rights and Fair Compensation

AI-generated outputs must clearly reference underlying data sources to respect intellectual property rights. Developers of commercial AI tools should financially compensate entities and individuals whose data they utilize. The distribution could be structured as follows:

- **Healthcare Data (e.g., Electronic Health Record or Clinical Trials data):**
 - Patients whose data is used: majority share (e.g., 90%)
 - Healthcare providers facilitating data and revenue exchange: modest share (e.g., 9%)
 - Governmental oversight and administrative costs: minimal share (e.g., 1%)
- **News or literary sources:**
 - Individual authors or original content creators: majority share (e.g., 90%)
 - Publishers or curators: moderate share (e.g., 9%)
 - Regulatory bodies: minimal share (e.g., 1%)

For instance, if an AI-generated book analysis references copyrighted literary works, the authors or rights holders should proportionally benefit. Table 1 presents an AI-based analysis of the Orson Card’s novel *Ender’s Game*. Had it been written by a generative AI system and commercialized, the revenues should have been shared with the copyright holders of the previous works.

Table 1: Relating to Earlier Works by Other Authors

Theme in	Earlier Work It Resembles	Connection
Military training as a social duty	<i>Starship Troopers</i> (Heinlein)	Heinlein advocates military citizenship; Card questions ethical implications.
Strategic deception	<i>The Art of War</i> (Sun Tzu)	Warfare as deception directly influences Ender's training methods.
Manipulation and societal conditioning	<i>Brave New World</i> (Huxley), <i>1984</i> (Orwell)	State manipulation parallels Battle School's controlled environment.
Gifted, isolated leader	<i>Dune</i> (Herbert)	Both Ender and Paul Atreides face heavy burdens imposed externally.
The psychological burden of command	<i>Crime and Punishment</i> (Dostoevsky)	Ender shares Raskolnikov's internal moral conflict after decisive, irreversible actions.
"Chosen One" narrative	<i>Hamlet</i> (Shakespeare)	Ender, like Hamlet, struggles under manipulation toward tragic outcomes.

An AI system using these works must explicitly cite them and, if applicable, distribute royalties proportionally.

Ecological Accountability

AI developers must internalize environmental costs, moving beyond current practices where ecological harms are externalized to society. Companies should implement measures like heat capture in data centers and recycling thermal energy for heating or power generation, thus minimizing overall carbon emissions.

Human-in-the-Loop Safeguards

Overreliance on AI in critical domains such as healthcare, aviation, and national security should be curtailed through strict regulations mandating human oversight. For example, Intelligent Clinical Decision-Making (ICDM) systems in healthcare should deliberately and periodically insert simulated errors unbeknownst to clinicians under controlled conditions and expect the clinician to respond with corrective feedback. If clinicians fail to detect these intentional inaccuracies, the system should alert and educate them to foster vigilance and ensure robust oversight of AI system performance. This reinforcement feedback process should never be applied in urgent and emergency care and the controlled conditions must ensure that patients cannot be harmed during this process.

Expanding AI Training Data

English is the predominant language on the Internet and for training AI systems, yet it represents only a portion of global cultural heritage. To maximize the capabilities of AI systems, it is essential to systematically translate legally accessible global knowledge resources (e.g., Chinese, Russian, French, Japanese, Arabic, etc.) into English datasets. For instance, translating non-English medical texts or ancient philosophical works significantly enriches training data and enhances AI's analytical depth.

Public AI Accessibility and Subscription-Based Redistribution

A portion of fees paid by subscribers using powerful AI models should be allocated to providing free AI services to the public using previous-generation models. While this approach is currently practiced by AI companies, it remains a fragile, profit-driven system that may not persist indefinitely. This arrangement should therefore be legislated to ensure continued public access to AI tools. At present, AI companies rely on free-tier users to improve their models through Reinforcement Learning from Human Feedback (RLHF). However, once this exchange ceases to be profitable, these companies, bound by fiduciary obligations to their shareholders, may discontinue free services. Unless legally required to serve the public interest, AI firms will likely prioritize short-term financial returns over long-term accessibility, making government intervention crucial in maintaining this system for public benefit.

Conclusions

Just as businesses must identify sustainable models to generate revenue, societies must establish frameworks to equitably distribute wealth and resources. As AI-driven automation increasingly replaces human labor with robots, a larger share of AI-generated revenues must be allocated to society at large.

Superintelligent software and robotic applications no longer belong to science fiction—they are becoming intrinsic to our near future. If we fail to establish a new economic and social framework before AI-driven automation erodes traditional labor markets, vast segments of society may find themselves economically obsolete. This is not a distant possibility but an imminent threat. If we continue to delay necessary policy interventions—just as we have done with climate change—we may be unable to prevent widespread societal collapse.

The AI revolution is built on centuries of human knowledge encoded in literature, science, philosophy, and the arts. This cultural heritage does not belong to a handful of corporations but to humanity as a whole. As AI harnesses this collective inheritance more effectively than ever, its benefits must be equitably distributed.

Superintelligent software and robotic applications represent imminent changes to societal structures, employment, and the economy. To avoid societal instability, we must proactively develop equitable economic models to ensure that the wealth generated by AI advancements benefits humanity broadly, not just corporate shareholders.

Currently, the United States lacks comprehensive federal legislation addressing these issues, potentially jeopardizing long-term stability. While some states have begun legislating on AI, comprehensive national standards are necessary for global leadership and societal resilience. The U.S. must balance national interests with ethical and cooperative international engagement, recognizing AI technologies as global public goods rather than proprietary national assets.

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Disclaimer

These recommendations reflect my individual perspective as an AI expert active since 1990, particularly in AI applications in medicine. Numeric examples are conceptual and require refinement through further expert analysis. Although I am a federal employee, my input is submitted as a private U.S. citizen due to procedural limitations.

References

1. State of Colorado, SB24-205 (2024).
2. State of California, AB-2013 (2024).
3. EU Artificial Intelligence Act, Article 53 (2024).
4. National Science Foundation, Federal Register Vol.90 (2025): 9088-9089.

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